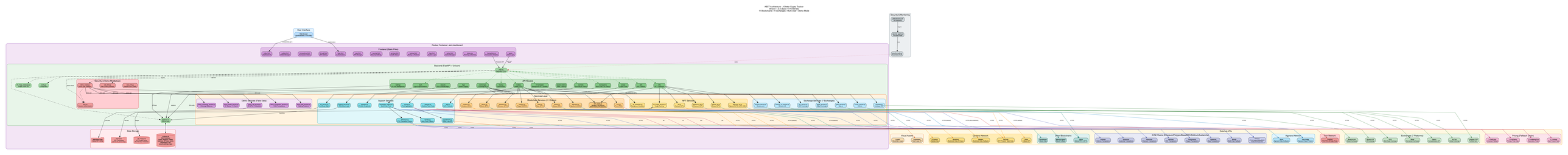

ABCT Architecture - A Better Crypto Tracker

Version 1.5.0 (Build 1770726755)

11 Blockchains | 7 Exchanges | Multi-User | Demo Mode

User Interface
Web Browser (localhost:8000 / Port 8080)
HTTP/HTTPS GET /
Authentication
Docker Container: abct-dashboard
Frontend (Static Files)
index.html Dashboard UI
wallets.html Wallet Manager
transactions.html Transaction History
nft-wall.html NFT Gallery
login.html Authentication
apis.html API Manager
security.html Security Settings
services.html Health Status
backup.html Backup & Restore
logs.html System Logs
cache.html Cache Manager
styles.css Theming (5 themes)
transactions.js Transaction Analytics
app.js Client Logic
Transaction API
REST API JSON
Backend (FastAPI + Uvicorn)
main.py Application Entry
global
rate limit
session check
scans
nft_image_database.py Image Cache DB
config.py Configuration
Security & Demo Middleware
demo_mode.py Demo User Detection
rate_limit.py Rate Limiting (slowapi)
size_limit.py Request Size (10MB)
auth.py Session Authentication
demo user
API Routers
/settings API Key Management
/auth Login/Logout/Sessions
/security SSL/Certificate Mgmt
/logs System Logging
/exchanges 7 Exchange APIs
/backup Export/Import
/wallets Wallet CRUD
/nft-scheduler Background NFT Updates
/custom-tokens Token Tracking
/portfolio Summary & History
/transactions Multi-Chain Txn History
/prices Crypto Prices
/defi DeFi Positions
/nfts NFT Collections
API keys
sessions
SSL config
Services Layer
Demo Services (Fake Data)
demo_exchange_service.py Exchange Balances
demo_wallet_service.py 91 Tokens, 11 Chains
demo_nft_service.py 76 NFTs, 7 Collections
demo_transaction_service.py 1,500 Transactions
demo_defi_service.py Staking Positions
Support Services
ssl_service.py Certificate Mgmt
logging_service.py Centralized Logs
snapshot.py Portfolio Snapshots
transaction_history.py Multi-Chain Txns
pricing.py Price Aggregation
taptools.py Cardano Token Data
defi.py DeFi Protocols
etherscan.py ETH Transaction History
moralis.py Multi-Chain Fallback
logostream.py Token Logo API
log entries
Blockchain Services (11 Chains)
cardano.py Cardano (ADA)
bitcoin.py Bitcoin (BTC)
ethereum.py Ethereum (ETH)
polygon.py Polygon (MATIC)
base.py Base (ETH)
solana.py Solana (SOL)
algorand.py Algorand (ALGO)
evm_chain.py Generic EVM Service (BSC/Arbitrum/Avalanche)
tron.py Tron (TRX)
NFT Services
nft_scheduler.py Background Updates
nft_image_service.py Image Caching
nft.py Cardano NFTs
ethereum_nft.py Ethereum NFTs
solana_nft.py Solana NFTs
algorand_nft.py Algorand NFTs (ARC-3/69)
BSC/ARB/AVAX NFTs
Exchange Services (7 Exchanges)
binance_service.py Binance.com
binance_us_service.py Binance.US
okx_service.py OKX Exchange
bitget_service.py Bitget Exchange
gate_service.py Gate.io
kucoin_service.py KuCoin
coinbase.py Coinbase CDP
database.py SQLite Layer
SVG files
cert files
images
Data Storage
static/demo-nfts/ Demo NFT SVGs
data/certs/ SSL Certificates (server.crt, server.key)
nft_images.db NFT Image Cache (thumbnails, metadata)
portfolio.db Main Database (users, sessions, wallets, balances, API keys, cache, transactions, nfts, security settings, logs)
HTTPS
HTTPS (fallback)
HTTPS (BSC/ARB/AVAX)
1st
2nd
3rd
4th
Security & Monitoring
.git/hooks/pre-push Git Integration
triggers
security_agent.py Pre-push Hook
runs
security_audit.py Automated Audit
External APIs
Visual Assets
LogoKit Blockchain Logos
Logostream Token Logo API
Cardano Network
CExplorer Staking Data
Blockfrost Blockchain Data (Primary)
Maestro Blockchain Data (Fallback)
TapTools NFT Prices & Token Data
Koios Fallback API
Other Blockchains
Blockstream Bitcoin Data
Mempool.space Bitcoin Fallback
Helius Solana RPC & NFTs
EVM Chains (Ethereum/Polygon/Base/BSC/Arbitrum/Avalanche)
Arbiscan Arbitrum Transactions
Snowscan Avalanche Transactions
Etherscan Ethereum Transactions
BscScan BSC Transactions
Polygonscan Polygon Transactions
Basescan Base Transactions
Moralis Multi-Chain Fallback
Alchemy ETH/Polygon/Base/BSC/Arbitrum/Avalanche
Algorand Network
Tatum Algorand Data (Fallback)
Pera Wallet Algorand Data (Primary)
Tron Network
TronGrid TRX/TRC-20 Data (Free)
Exchanges (7 Platforms)
Binance.com Global Exchange
Binance.US US Exchange
OKX Multi-Asset Exchange
Bitget Derivatives Exchange
Gate.io Comprehensive API
KuCoin Global Exchange
Coinbase CDP Portfolio Sync
Pricing (Fallback Chain)
4. DefiLlama Universal Fallback
1. CoinGecko Primary Price Data
2. CoinMarketCap Secondary Prices
3. Coinbase Spot Prices (Free)